

Victoria Simmons

ML FINAL EXAM

This final project challenged me to build a complete machine-learning pipeline using a real Airbnb dataset. I worked through data cleaning, feature engineering, model training, evaluation, and ensemble learning. Seeing the transformation from raw, messy data to a fully trained predictive model helped me understand how each step in the ML workflow contributes to the final outcome.

Data Cleaning & Preparation

One of the first things I learned is how essential data cleaning is. The raw dataset had inconsistent formatting, missing values, and messy price fields. For example, the notebook shows how price and service fees were stripped of symbols and converted to numeric values:

```
“df[c] = df[c].astype(str).str.replace('$', '', regex=False).str.replace(',', '', regex=False)”
```

I also handled missing values using medians for numeric columns and modes for categorical ones. Fixing typos like "brookln" and "manhatan" in the neighborhood column reminded me how small errors can affect model performance:

```
“df['neighbourhood_group'] =  
df['neighbourhood_group'].fillna('Unknown').replace({'brookln':'Brooklyn','manhatan':'Manhattan'})”
```

This part of the project taught me that clean data is the foundation of any reliable model.

Feature Engineering

Creating new features helped me understand how additional information can improve

predictions. For example, I engineered a fee-to-price ratio:

```
“df['fee_to_price_ratio'] = df['service_fee'] / df['price']”
```

I also scaled construction years and log-transformed review counts to reduce skew:

```
“df['reviews_log'] = np.log1p(df['number_of_reviews'])”
```

These transformations made the data more suitable for machine-learning algorithms.

Model Training

Training multiple models was one of the most rewarding parts of the assignment. I trained:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- KNN
- SVC

The notebook shows each model being fit:

```
“model.fit(X_train, y_train)”
```

Comparing accuracy, precision, recall, F1-score, and ROC-AUC helped me understand how different algorithms behave with the same dataset.

Ensemble Learning

The most advanced part of the project was building two ensemble models:

Soft Voting Ensemble

I selected the top three models based on F1-score:

```
“top3_names = results_df['Model'].head(3).tolist()”
```

Bayesian Accuracy-Weighted Ensemble

This ensemble averaged model probabilities using accuracy-based weights:

```
“return np.average(probs, axis=0, weights=weights)”
```

This taught me how combining models can produce more stable and reliable predictions.

Challenges I Overcame – Cleaning inconsistent price formats. Handling missing values.

Balancing the dataset using downsampling. Understanding how categorical encoding affects model behavior. Comparing multiple models without getting overwhelmed. Learning how ensemble methods improve performance. Each challenge pushed me to think critically and apply what I learned throughout the course.

What I Learned – This project helped me understand the full lifecycle of a machine learning

workflow: Data cleaning is not optional. I’ve learned it’s essential. Feature engineering can dramatically improve model performance. No single model is “best” for every dataset.

Ensembles often outperform individual models. Visualizations make results easier to interpret.

GitHub is an important part of sharing and documenting work.

Conclusion– Completing this final exam gave me confidence in my ability to build and evaluate machine-learning models from scratch. Uploading my notebook to GitHub allowed me to present my work professionally and clearly. This project tied together everything I learned in the course,

and I'm proud of the pipeline I built.